

Milestone-04: Incident Type Analysis

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1. Project Description

Milestone-4 focuses on analyzing the distribution of disaster incident types and understanding their relationship with geographic regions and assistance programs. The cleaned dataset from previous milestones is used to perform incident-level analysis and derive meaningful insights. Disaster records are aggregated based on incident types and states to identify the most frequent disasters and region-specific patterns. Additionally, the relationship between disaster types and assistance programs is analyzed to understand their impact. This milestone integrates insights from temporal and geographical analysis to provide a comprehensive view of disaster trends.

2. Dataset Used

The dataset used for this milestone is **usnd_cleaned.csv**, which was generated during **Milestone-1** after cleaning the original disaster dataset. It contains records of natural disaster declarations across different states in the United States. The dataset includes key attributes such as **incident type, state, year, month, and assistance program indicators such as Individual Housing(IH) and Public Assistance(PA)**. The dataset was validated to ensure consistency and accuracy, making it suitable for incident type analysis and visualization.

3. Environment Setup

The analysis was performed using Python in a development environment such as Jupyter Notebook or VS Code. Libraries such as Pandas and NumPy were used for data processing, while Matplotlib and Seaborn were used for visualization. These tools enabled efficient aggregation, analysis, and graphical representation of disaster data based on incident types.

4. Data Exploration

Before performing incident type analysis, the dataset was explored to understand its structure and verify data quality.

4.1 Loading the Dataset

The dataset was loaded using Pandas to begin the analysis process.

4.2 Inspecting Dataset Structure

The dataset structure was examined to verify column names and data types, ensuring the presence of required attributes such as incident type, state, and assistance indicators.

4.3 Examining Incident Types

Different disaster categories such as floods, storms, hurricanes, and fires were identified to understand the variety of disasters included.

4.4 Checking Missing Values

The dataset was checked for missing or invalid values, particularly in incident type and assistance-related columns, to ensure accurate analysis.

5. Data Preprocessing

Data preprocessing is a crucial step in the analysis pipeline, as it ensures that the dataset is clean, consistent, and suitable for further analysis. In this stage, the disaster dataset is carefully prepared by validating data quality, handling missing values, and organizing the data for aggregation and visualization. Proper preprocessing improves the accuracy and reliability of the insights derived from the analysis.

5.1 Dataset Loading

The cleaned dataset (*usnd_cleaned.csv*) was loaded into the analysis environment using the Pandas library. This step enables access to the dataset for further processing and analysis.

Purpose:

- To import the dataset into the working environment
- To verify that the dataset is correctly loaded

Steps:

1. Import necessary libraries such as Pandas
2. Load the dataset using `pd.read_csv()`
3. Display the first few rows to confirm successful loading

5.2 Feature Verification

After loading the dataset, the next step is to verify the availability and correctness of important features required for analysis. Key columns such as *incident type*, *state*, *year*, *month*, and *assistance program indicators* were checked to ensure they exist and are properly formatted.

Purpose:

- To confirm the presence of required columns
- To ensure column names are consistent and usable

Steps:

1. Display all column names using `df.columns`
2. Check data types of each column
3. Rename columns if necessary for consistency

5.3 Handling Missing Values

The dataset was examined for missing or invalid values, especially in critical columns such as incident type, state, and assistance indicators. Missing data can affect analysis results, so appropriate steps were taken to handle such values.

Purpose:

- To maintain data accuracy and consistency

- To avoid errors during aggregation and visualization

Steps:

1. Identify missing values using `df.isnull().sum()`
2. Analyze the extent of missing data
3. Remove or handle missing values using methods like `dropna()` or filling techniques
4. Ensure that key columns do not contain null values

5.4 Data Preparation

Once the dataset was cleaned and validated, it was prepared for analysis by organizing it for aggregation. The data was structured to enable grouping based on incident types and states, which is essential for generating insights and visualizations.

Purpose:

- To organize data for efficient analysis
- To prepare the dataset for grouping and visualization

Steps:

1. Ensure all required columns are present and cleaned
2. Convert data into appropriate formats if needed
3. Prepare the dataset for grouping operations such as:
 - Incident type-wise aggregation
 - State-wise aggregation
4. Finalize the dataset for visualization and analysis

6. Proposed System

The proposed system focuses on analyzing disaster incident types and their impact across different states.

6.1 Incident Type Distribution

The dataset is grouped by incident type to calculate the frequency of each disaster category.

6.2 State vs Incident Analysis

The dataset is grouped by state and incident type to identify which disasters are dominant in specific regions.

6.3 Assistance Program Analysis

The relationship between incident types and assistance programs is analyzed to understand how different disasters trigger support mechanisms.

6.4 Visualization

Various visualizations are generated, including:

- Bar charts for incident distribution
- Heatmaps for state vs incident type
- Grouped bar charts for assistance programs

6.5 Insight Generation

The system generates insights by interpreting the visualizations to identify patterns, dominant disaster types, and their impact.

7. Training Pipeline

The pipeline includes several key features:

- **Incident Aggregation** – Grouping data by incident type and state
- **Comparative Analysis** – Comparing disaster types across regions
- **Visualization Generation** – Creating charts and heatmaps
- **Impact Analysis** – Studying assistance program usage
- **Insight Extraction** – Identifying key patterns and trends

This structured pipeline ensures effective analysis of disaster incident data.

8. Conclusion

In this milestone, disaster incident types were analyzed to understand their distribution and impact across different states. The dataset was processed and aggregated to identify the most frequent disaster types and their regional variations. The relationship between disaster types and assistance programs was also examined to assess their impact on communities.

Visualizations such as bar charts and heatmaps helped in identifying dominant disaster types and affected regions. The analysis provides valuable insights into disaster patterns and supports better decision-making for disaster management and preparedness.

Dashboard : <https://disaster-dashboard-3byw3chon-dhruthi-paritala-s-projects.vercel.app>